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Bachelor's thesis

Modeling Time-Varying Short-Term Dependence in Foreign Exchange Time Series

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Abstract

Financial time series are well known to exhibit non-stationary behaviour and structural change. These structural changes can influence investment strategies and planning. This text presents and explains data from foreign exchange markets, and provides a description of autoregressive models.

1 Introduction

sec:intro

The analysis of financial time series is an important field in econometrics and blablabla.

2 Background

sec:background

2.1 Stationarity

ec:stationarity

Stationarity is a central concept in time series analysis. The literature distinguishes between *strict* and *weak* stationarity. For most applications, weak stationarity is of interest, see Tsay Tsay 2005, p. 25.

Definition 2.1. A time series $\{r_t\}$ is weakly stationary if both the mean of r_t and the covariance between r_t and r_{t-l} are time-invariant, where l is an arbitrary integer.

3 Foreign Exchange Data

sec:FX

The foreign exchange (FX) tick-by-tick data used in this report is obtained from https://www.histdata.com/, which provides freely available tick and intraday FX data. In this report we consider the EUR/USD exchange rate. Each data entry consists of a Date-Time stamp, bid quote, ask quote, and volume. For instance, a data entry may look like this:

```
20260101 170401135,1.173870,1.175320,0
```

This entry corresponds to a bid-price of 1.173870 and an ask-price of 1.175320 on January 1st 2026 at 17:04:01,135.

Let B_τ and A_τ denote the bid and ask prices at tick time τ. Following common practice in high-frequency econometrics, I use the mid-price

$$P_\tau = \frac{B_\tau + A_\tau}{2}$$

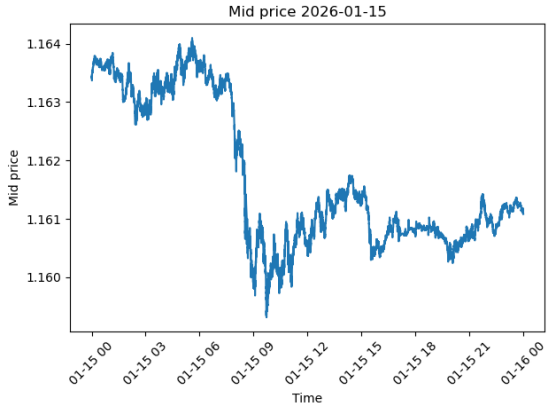

Figure 1: EUR/USD mid-price on January 15th 2026. The market is different at the start and end of the day.

fig:midprice

as a proxy for the transaction price. Figure 1 shows an example of the EUR/USD mid-price over a single trading day, constructed in this way.

Note that the market is very different at the start and end of the day. We may therefore clip the each day, and only consider data from the main New York and London trading hours, which in our case means restricting the sample to the combined window 03:00-16:00 Eastern Standard Time (EST).

3.1 Pre-Averaging and Returns

subsec:preavg

Let P_τ denote the observed transaction price at tick time τ. The observed price can be decomposed as

$$P_\tau = P_\tau^* + \eta_\tau,$$

where P_τ^* is the latent efficient price and η_τ represent the microstructure noise. To mitigate the impact of noise, we apply pre-averaging as discussed in Mykland and Zhang 2016. In practice, we consider a local averaging scheme over k consecutive ticks. The pre-averaged price is defined as

$$\overline{P}_t = \frac{1}{k}\sum_{j=0}^{k-1} P_{t-j},$$

where t now indexes the last tick in each averaging block. Returns are then constructed as

$$r_t = \overline{P}_t - \overline{P}_{t-1}$$

4 Autoregressive Models

sec:AR

An autoregressive model of order one (AR(1)) for a time series $\{r_t\}$ is given by

$$r_t = \mu + \phi r_{t-1} + \epsilon_t,$$

where μ is the unconditional mean return, ϕ measures the linear serial dependence and ϵ_t is a normally distributed innovation term with mean 0 and variance σ^2. As discussed in Tsay 2005, p. 33, weak stationarity is both a sufficient and necessary condition for AR(1) models.

sec:discussion

5 Discussion

sec:plan

6 Plan for the Project

cknowledgements

7 Acknowledgements

References

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Tsay

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